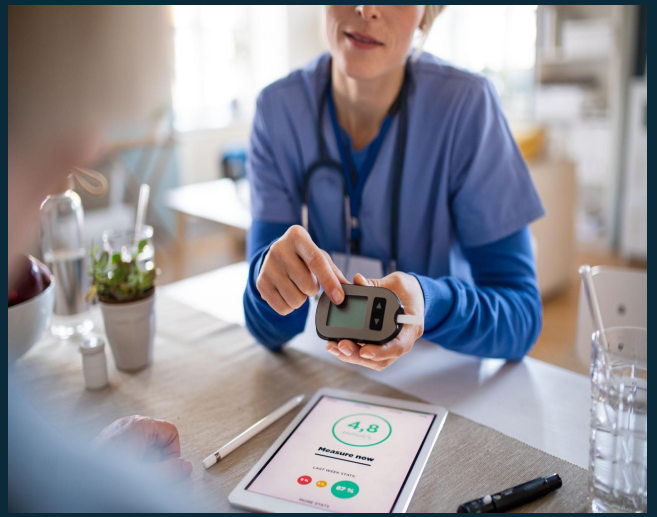
GlucoTwin

An explainable, personalized glucose Digital Twin for type 2 diabetes care

Digital Twin Challenge 2026

Team Name : Coffee & Panic
Institution : Marathwada Mitra Mandal's College of Engineering , Pune
Team Members : Jidnyasa Patil , Sajiree Damle



All demonstration patients and all reported results are SYNTHETIC. Not clinically validated. Not a medical device.

Healthcare problem: diabetes at national scale

Type 2 diabetes dominates; day-to-day glucose between clinic visits is largely invisible

~101 million

people with diabetes in India (11.4% of the population), ICMR-INDIAB estimate

~136 million

people with prediabetes (15.3%), a pool that can convert to diabetes

113,043

survey participants across states and UTs, 2008-2020

Why forecasting + a twin?

Continuous glucose monitors produce a reading every few minutes, but clinicians see snapshots. A continuously updated, explainable patient state with honest uncertainty could help a clinician review where a patient is heading. This is our motivation, not a demonstrated benefit.

Source: Anjana et al., Lancet Diabetes Endocrinol 2023 (DOI 10.1016/S2213-8587(23)00119-5), figures as reported by secondary coverage and a Natl Med J India commentary; verify against the paper.

Existing solutions and literature

What we found - and what we could not verify

Digital twins for health

Scoping reviews (e.g. Katsoulakis et al., npj Digit Med 2024) survey health digital twins and their applications. We use the working idea of a personalised, continuously updated, predictive patient model.

CGM forecasting

Published work forecasts glucose from CGM history (e.g. GARNN, arXiv 2402.16230; a Bayesian T2DM approach, arXiv 2409.07315). We did not reproduce them; we benchmark only against persistence and ridge.

Open datasets

DiaTrend: type 1, controlled access. ShanghaiT2DM: type 2, 15-min CGM, public but licence terms conflict across sources. Glucose-ML (2025) surveys these collections.

Other challenge entries

Public repositories exist for this challenge, including one already named 'GlucoTwin' and one 'ThaliTwin'. We did not audit their claims; we flag the name conflict.

Research gaps and our contribution

A small, carefully evaluated proof of concept

Gaps we target

Leakage-resistant, patient-level evaluation against simple baselines

Explicit behaviour for stale, missing and corrupt input (abstain, not guess)

Explanations that are faithful to the model, with causal limits stated

What-if tools that do not invent physiology

What this repository contributes

Twin state engine: validation, quarantine, forecast history, audit, restart-safe

Per-horizon forecasts (15-120 min) with conformal intervals

Exact additive explanations and revision tracking

Scenario lab limited to supported inputs

54 tests; reproducible results with ablations reported even when negative

Problem statement and intended outcome

One localised outcome, not the whole body

Primary task

Given the last 60 minutes of CGM readings (15-min grid) plus patient history, forecast glucose for the next 15-120 minutes with an uncertainty interval, an explanation and a data-quality status.

Intended outcome

A working data-to-forecast loop that a clinician-facing dashboard can show, test and audit - and an honest benchmark.

Explicitly out of scope

Diagnosis, treatment or insulin/medication advice. Alarms. Any claim of clinical validation.

Data sources, privacy and limitations

Research found hard limits; we documented a safe fallback

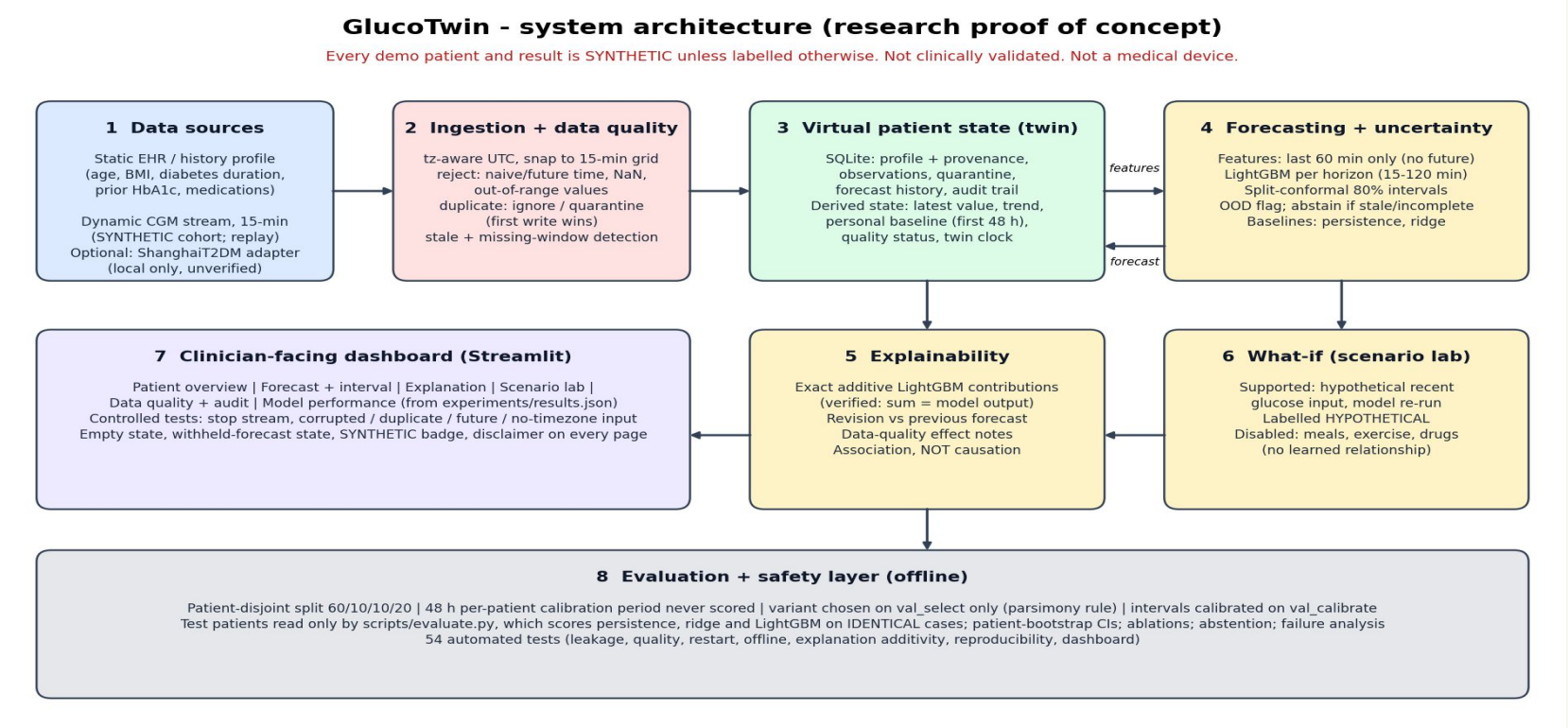
Dataset	Population	Access finding	Used?
DiaTrend	Type 1, CGM	Controlled access (Synapse + agreement); no redistribution	No
ShanghaiT2DM	Type 2, 15-min CGM	Public, but licence terms conflict across sources; host unreachable from the build sandbox	Adapter only (untested on real files)
Synthetic cohort	T2D-like, 100 pts x 14 d	Generated here, seeded, MIT, labelled SYNTHETIC on every record	Yes - all results

- No real patient data in the repository, logs or figures
- Static-to-glucose links in the synthetic data exist by construction: they cannot validate physiology
- Real-data results must be produced and reported separately

No result in this deck says anything about real patients.

Architecture and data fusion

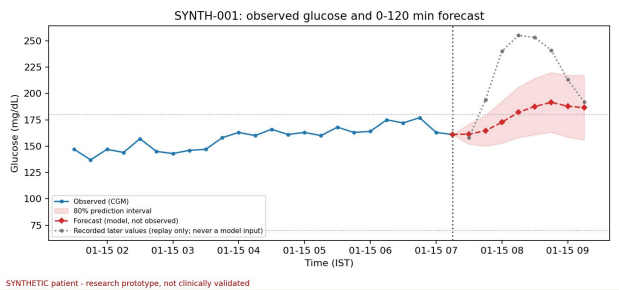
History + stream -> quality gate -> twin state -> forecast -> explanation -> dashboard

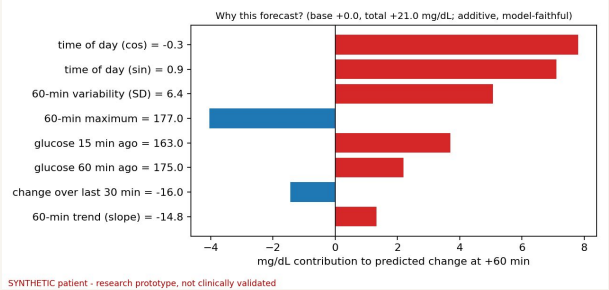


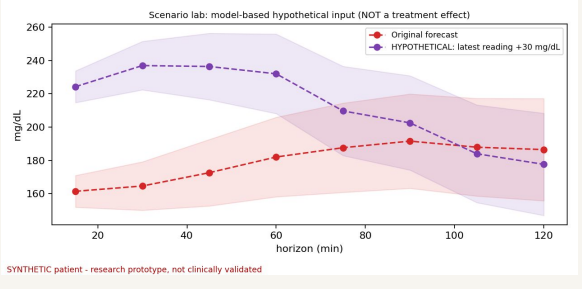
Dynamic update workflow and dashboard

Every new reading updates state, forecast, explanation and audit trail

1. **New reading**
2. **Validate**
3. **Append / quarantine**
4. **Quality check**
5. **Forecast or withhold**
6. **Explain + audit**



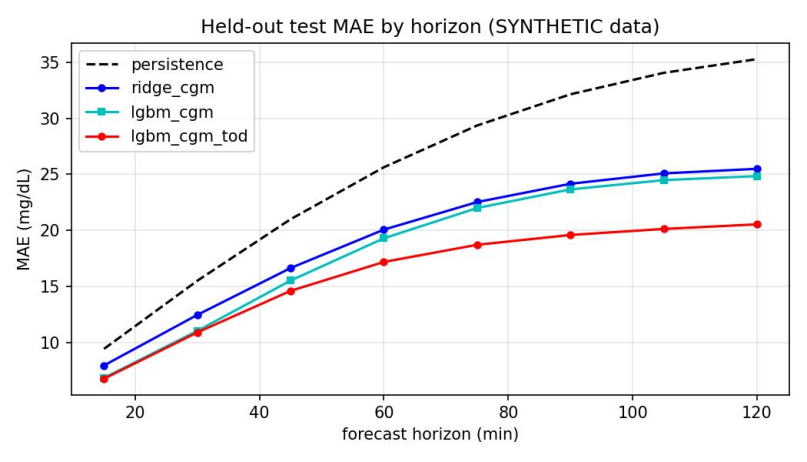




- Six tabs: overview, forecast, explanation, scenario lab, quality + audit, model performance
- Explanations are exact additive terms of the model output; scenario lab is labelled HYPOTHETICAL, meal/exercise/drug controls disabled
- Replay of a recorded SYNTHETIC stream; there is no live-device connection

Model evaluation and baseline comparison

Held-out SYNTHETIC test: 20 patients, 20,825 windows, identical cases for all models



MAE mg/dL	+15 min	+30 min	+60 min	+120 min
Persistence	9.5	15.5	25.6	35.3
Ridge	8.0	12.5	20.1	25.5
LightGBM (CGM only)	6.9	11.1	19.3	24.8
LightGBM + time of day	**6.8**	**10.9**	**17.2**	**20.5**

- Primary vs persistence at +60 min: -8.44 [-11.12, -6.24] mg/dL (95% patient-bootstrap CI); better for 20 of 20 patients
- Ablations vs CGM-only at +60 min: time of day -2.11 [-2.70, -1.47] (expected by construction: synthetic meals are scheduled); personal baseline -0.05 [-1.02, 1.04]; static EHR +0.05 [-0.73, 0.86] - no reliable gain at +60 min (personal baseline helps only ~0.3 mg/dL at +15 min)

Synthetic data: shows the pipeline works and recovers the generator's dynamics - not clinical accuracy. Preliminary.

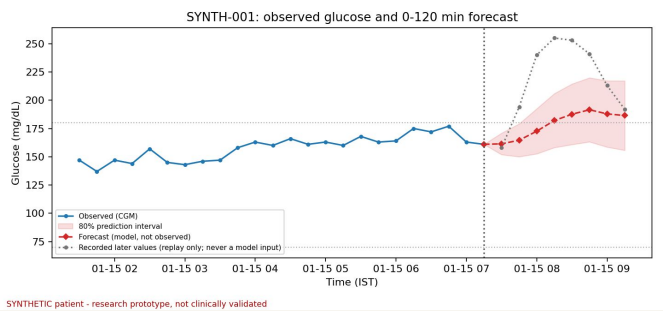
Tests, data-quality behaviour and uncertainty

Fail safe, show uncertainty, admit misses

FORECAST WITHHELD

newest reading is 45 min old (limit 20 min): stream stale

No false precision: stale or incomplete input is never imputed.



SYNTHETIC patient - research prototype, not clinically validated

- Stale (> 20 min) or incomplete 60-min window: forecast withheld; 6.4% of observed test origins (synthetic missingness)
- Rejected / quarantined: naive or future timestamps, NaN, out-of-range values, conflicting duplicates
- 80% intervals: coverage 78-80% on test; per patient 71-85% (marginal guarantee only)
- Hardest case: flat glucose before a meal (MAE 22.6 vs 12.1 mg/dL rising, +120 min); bottom-left: a breakfast spike escaping the interval
- 54 automated tests incl. leakage, restart, offline, additivity, dashboard; mutation-checked

Limitations, responsible use and next steps

Not claimed

Clinical validation or real-patient accuracy

Benefit for type 2 populations (data are synthetic)

Causal effects of food, exercise or drugs

Use for diagnosis, treatment or medication decisions

That time-of-day or EHR gains transfer to real data

Next steps

1. Run the ShanghaiT2DM adapter locally; verify columns, units, licence
2. Re-train and evaluate; report real-data results separately
3. Add meal/activity inputs only if real data support them
4. Clinician review and prospective evaluation before any clinical use
5. Resolve the project-name conflict

Conclusion

- A reproducible twin pipeline: fused history + stream, versioned forecasts, explanations, safe withholding
- On held-out synthetic patients it beats persistence and ridge; value on real patients is untested
- Negative and null findings are reported alongside positives

Repository: https://github.com/Jidnyasa-P/GlucoTwin
Demo video (unlisted): https://youtu.be/1czyEqa0aPc
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Research proof of concept. Synthetic data. Not clinically validated. Not a medical device.